Student analytics dashboard

- ❖ Project Type : Data Analytics Project
- ❖ Tools Used : Excel, PostgreSQL, Power BI
- ❖ Dataset : Student Performance — 5000 Records

Objective

- ★ The objective of this project is to analyze student academic performance, attendance, examination marks, results, and fee status using Excel, PostgreSQL, and Power BI. The project aims to identify performance trends and present meaningful insights through an interactive dashboard.

Tools Used

- ❖ Microsoft Excel – Data cleaning and preparation
- ❖ PostgreSQL – Data storage and SQL analysis
- ❖ Power BI – Interactive dashboard and data visualization

Dataset Description

- ★ The dataset contains 5,000 student performance records with details such as student information, attendance, assignment scores, internal marks, exam marks, total marks, grades, results, academic year, and fee status.

Data Preparation

- ★ The dataset was cleaned and prepared using Microsoft Excel. Duplicate records, data types, ranges, and categorical values were checked to ensure the data was suitable for analysis.

PostgreSQL Analysis

- ★ The cleaned dataset was imported into PostgreSQL for structured data analysis. SQL queries were used to analyze class-wise performance, subject-wise performance, pass and fail percentages, gender-wise performance, grade distribution, academic year trends, fee status, and top and bottom performing students.

Power BI Dashboard

- ★ Power BI was used to create an interactive dashboard to visualize student performance. The dashboard includes KPI cards, charts, slicers, and performance comparisons for better understanding of the data

Key Insights

- ★ The analysis helped identify differences in student performance across classes, subjects, genders, and academic years. It also provided insights into pass and fail rates, attendance patterns, grade distribution, fee status, and the highest and lowest performing students.

Conclusion

- ★ This project demonstrates how Excel, PostgreSQL, and Power BI can be used together to transform raw student data into meaningful insights. The interactive dashboard makes it easier to understand student performance and supports data-driven decision making

Project Outcome / Learning

- ★ Through this project, I gained practical experience in data cleaning, SQL analysis, data visualization, dashboard creation, and extracting meaningful insights from a large dataset. I also learned how different data analytics tools can be combined to complete an end-to-end analytics project.